COMP1801 - Machine Learning Coursework Report

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Word Count: 2975 words

1. Executive Summary

This report outlines a machine learning service which can be used by a manufacturing company to determine whether new metal alloy parts have a reasonable lifespan of operation. Since the old-fashioned lifespan testing is slow and non-destructive, the company needs a data-based process of estimating the precise lifespan of a component, as well as whether it has reached a minimum of 1500 hours of safety or not. Alloy composition measures, defect measures and processing parameter measures were applied using the given dataset of processing parameters to implement and test various regression and classification models.

To make the lifespan prediction, linear regression, Elastic Net, and Random Forest Regressors were created. Random Forest has the best performance with the highest R2 value of about 0.94, and it can fit the complex and non-linear relationship that can be observed in the data. In the task of classification, a comparison was made between the Logistic Regression and a Random Forest Classifier, where the latter showed better performance in terms of accuracy, ROC-AUC and precision-recall properties.

In general, the Random Forest models can be suggested to be deployed, as it give a reliable estimation of lifespan and a strong detectability of defective parts.

2. Data Exploration

The data have initially been loaded into a pandas data frame, where the total number of observations was 1000, and the total number of features was 16, without missing or repeating data, as the df.info and df.isnull results of the collab notebook can confirm.

This data consists of both numeric processing values (e.g. cooling rate, quench time, nickel percentage, defect counts, etc.), and non-categorical manufacturing information (type of part, type of microstructure, type of cast, location of seed, etc.). Lifespan is the target

variable, which measures the durability of each component of the metal (Stirling et al., 2021).

Descriptive statistics were used to point out a mean lifespan of about 1282 hours, with minimum and maximum lifespans of 359 and 2046 hours, respectively, which demonstrates a large range in the quality of the products. Some continuous data also exhibited wide ranges indicating possible non-linear or interaction effects. The heatmap of correlations (page 7) showed that most of the numerical features have low linear relationships with lifespan, whereas Nickel% associates with lifespan in a positive manner, and Iron% associates with it with a moderately negative relationship.

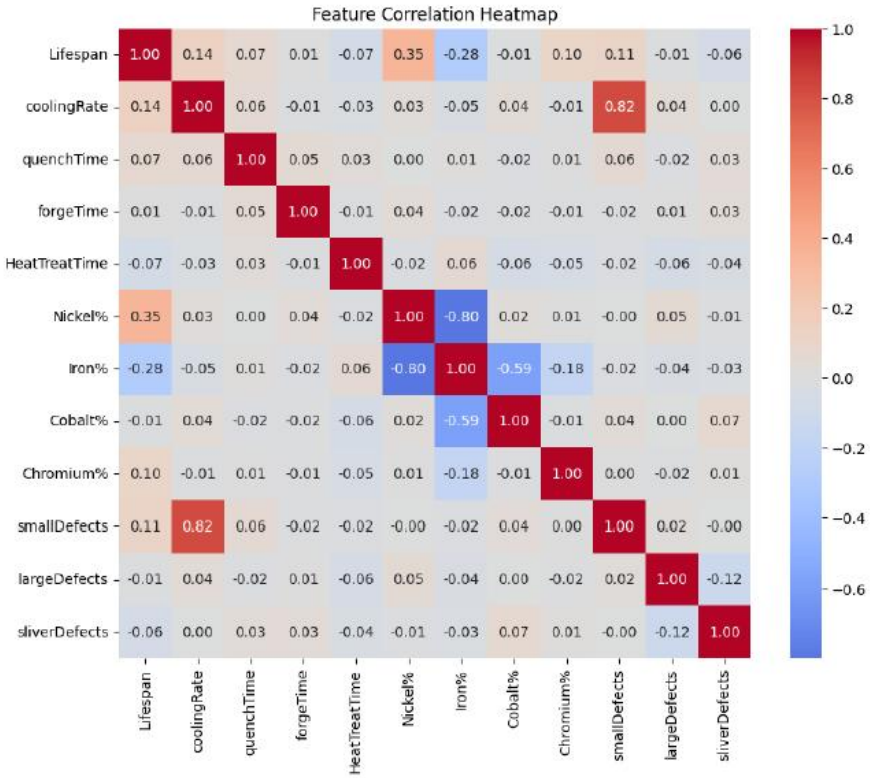


Figure 1: Heat map (Colab Notebook File, p7)

These relationships were also supported by scatterplots of these variables versus lifespan (pages 13-14), which indicated that there were upward and downward trends, respectively.

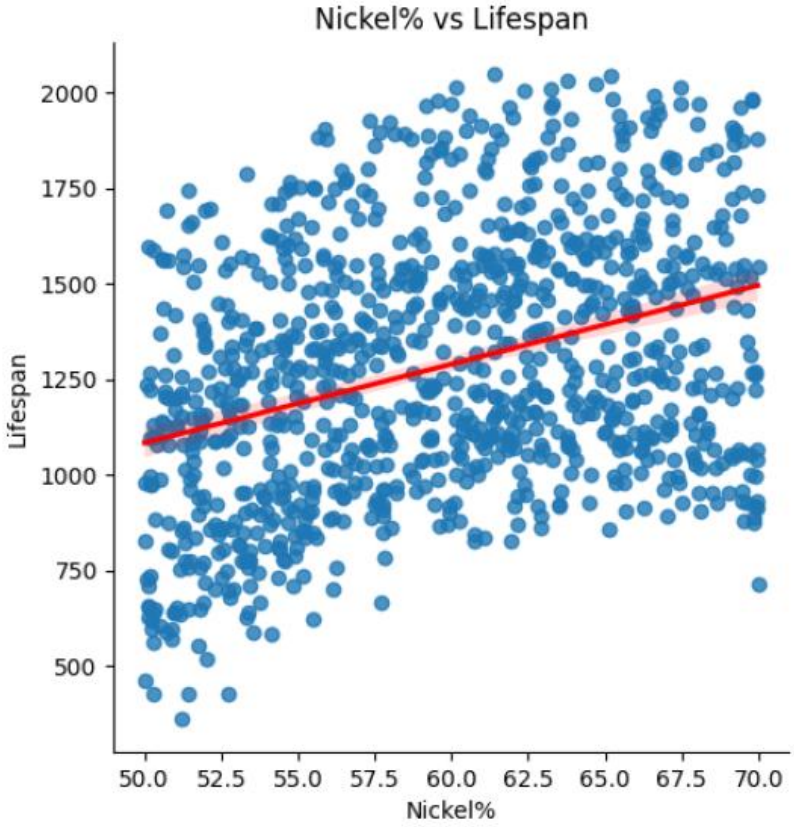


Figure 2: Nickel vs. Lifespan (Colab Notebook File, p13)

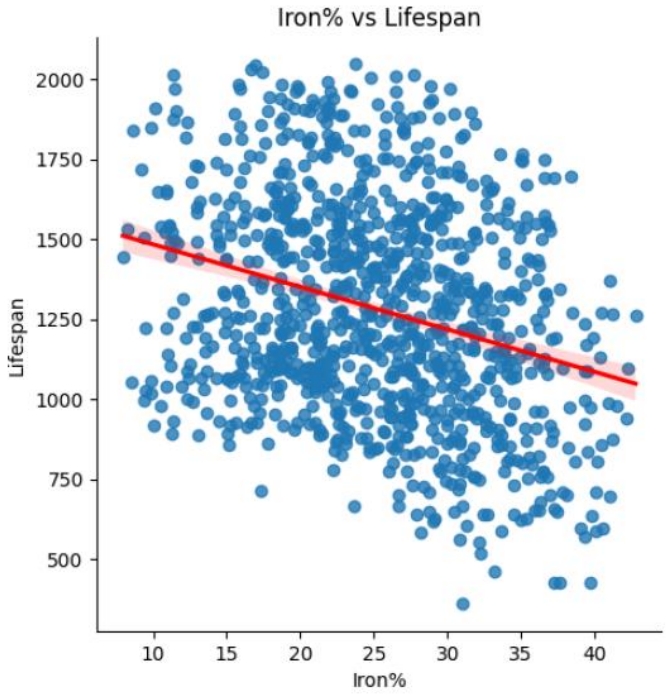


Figure 3: Iron vs. Lifespan (Colab Notebook File, p14)

The sample analyses of the categorical variables (pages 19-23) reveal that there are some observable differences in median lifespans by part type and by casting technique and therefore the categories of manufacturing processes could have a strong effect on the durability.

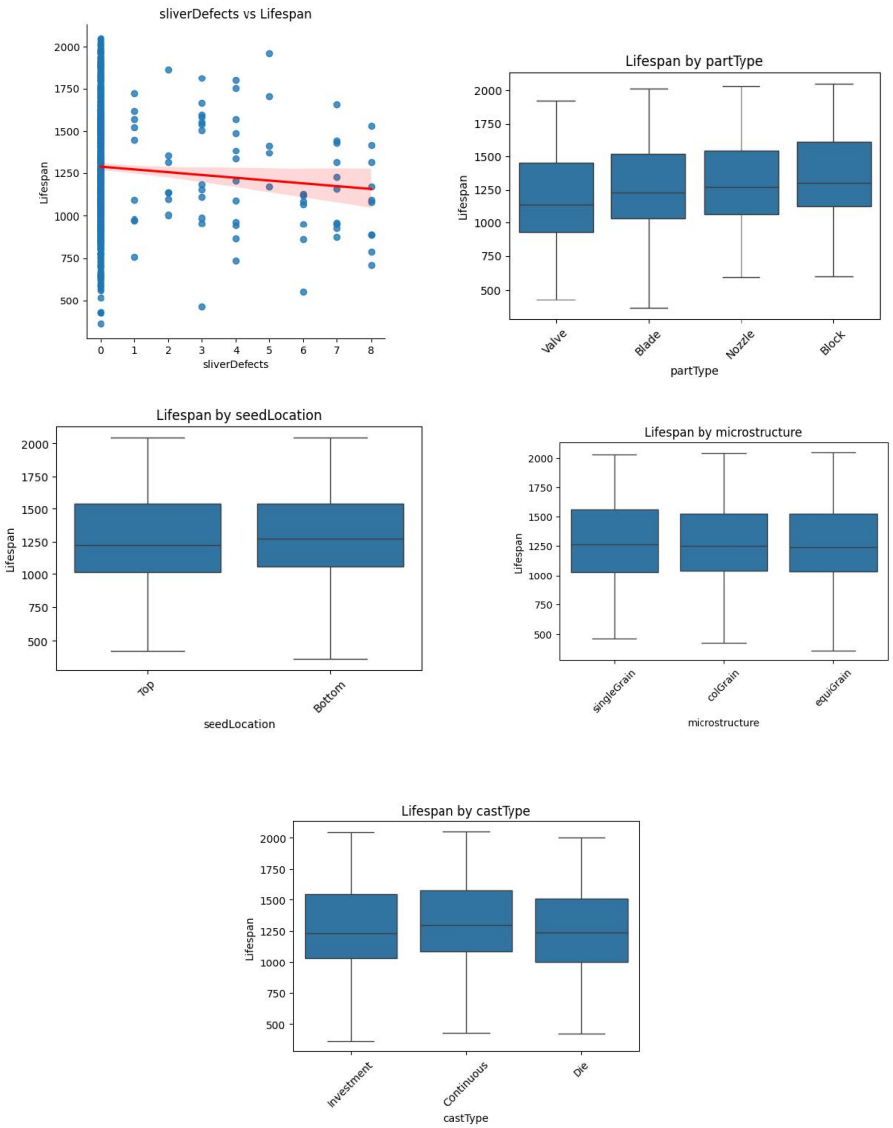


Figure 4: Differences in Median Lifespans (Colab Notebook File, p19-23)

Numeric features are represented by histograms (page 7) that indicate approximately uniform distributions or even mild skewness distributions, indicating that standardisation will prove useful in subsequent modelling.

Based on these observations, variables such as variables of alloy composition (Nickel%, Iron %), thermal processing (cooling Rate, HeatTreatTime) and selected categorical variables seem to emerge as the most relevant predictors. The fact that non-linear patterns

are visible as well as the interactions across the trees imply that the tree based or ensemble models could be better than solely linear models (Rosen et al., 2022). Thus, regression and classification are both likely to be benefited by a model with the ability to model non-linear behaviour, e.g. Random Forests.

3. Regression Implementation

3.1 Methodology

To estimate the life of the metal components manufactured, two families of regression models were chosen: linear regression, Elastic Net, and Random Forest Regressor. Linear models are an interpretable baseline which are appropriate in identifying simple additive relationships between processing parameters and lifespan. Non-linear trends were however, evident in the Part 2 data exploration, especially in the correlation between Nickel%, Iron%, and coolingRate. This fact is why a more adaptable model such as Random Forest was chosen as well since this tool can inherently record non-linear interactions and feature dependencies without the need to assume a linear structure (Chaabene, Flah and Nehdi, 2020). Elastic Net was also added as a regularised linear model that could deal with multicollinearity in the alloy composition features. A combination of these three models offers a comparative balance between simple, regularised and non-linear models.

The pipeline was made consistent with the help of scikit-learn ColumnTransformer and Pipeline (comp1801 code pdf, cells 118-124). All the numerical attributes (e.g., coolingRate, Nickel%, HeatTreatTime) were normalised through StandardScaler so that varying magnitudes of the features should not have a disproportionate effect on the linear models. OneHotEncoder was used to encode categorical variables (part type, microstructure, seedLocation, casttype), meaning that they are represented as binary indicator variables without ordinality. This pre-processing phase was uniform and guaranteed the fairness of all model assessments and also facilitated easy hyperparameter optimization.

The data set was divided into 80% training and 20% testing with random state=42 to ensure reproducibility. Three complementary regression measures, namely R2, which measures the explained variance; RMSE, which punishes bigger errors and is suitable when using continuous manufacturing tolerances; and MAE, which gives a more interpretable measure of average absolute error, were used to evaluate model performance (Srinivasan and

Deepalakshmi, 2022). A basic baseline model, which predicts the mean training lifespan, showed an R2 of about -0.037, which set a performance lower limit of future models.

GridSearchCV was used to carry out hyperparameter tuning. In the case of Elastic Net, tuning of alpha (the strength of overall regularisation) and l1ratio (the weight between L1 and L2 regularisation) took place because they directly affect the sparsity and robustness of the model. In the case of the Random Forest, nestimators (number of trees), maxdepth (tree complexity) and minsamplessplit (minimum samples to split a node) were defined as grids.

3.2 Evaluation

The process of evaluation aimed at the determination of the best regression model to predict the part lifespan by comparing the performance of Linear Regression, ElasticNet, and the Random Forest Regressor obtained after implementation of consistent pre-processing and the use of hyper parameters. All models had been trained on an equal 80/20 train-test split to be able to compare them fairly, and evaluated on the basis of R2, RMSE, and MAE, which allows assessing both the extent of variance explained and the magnitude of the error. A naive predictor, which attributed the average training lifespan to all samples, had an R2 of -0.037 and an RMSE of around 351 hours (comp1801 code pdf), and demonstrated that any influential model should be able to perform better than this naive one.

Linear Regression Model: The linear regression model showed a slight improvement on the base case with R 2 of 0.169, RMSE of 314.4 and MAE of 271.4. These findings show that the model could partly fit in the linear structure of the data, but it was highly constrained by the loose linear associations that arose when exploring the data. Tuned to several alpha and l1ratio values, the ElasticNet model had similar performance with 0.161 R2 and 315.8 RMSE. The optimised hyperparameters (alpha = 1.0, l1ratio = 0.9) suggest that the model is biased towards a heavy regularisation and a strong L1 penalty is used. Though this assisted in controlling multicollinearity in such features as Nickel% and Iron, the small increase in performance over Linear Regression highlights the low ability of linear processes to model intricate interaction.

Random Forest Regressor: The Random Forest Regressor, in its turn, has shown a significant increase in performance with an R2 of 0.943, an RMSE of about 82.15 and an MAE of 64.64, which is much higher than both the linear models and the baseline. The

process used to optimise the hyperparameters showed that a higher number of estimators and deeper trees, as hyperparameters, enhanced predictive accuracy, which was the capacity of the model to describe non-linearities and high-order interactions among processing parameters (Idowu, Fakoya and Olugbebi, 2025). The dramatic reduction in RMSE indicates the greater predictive power of Random Forest to reduce huge prediction errors, which is an important necessity in predicting the lifespan within a high-stakes manufacturing environment.

Comparing directly the last tuned models, it is evident that the best architecture to use in this regression problem was Random Forest. Unlike linear models, which were limited by the nature of their functional form and were incapable of capturing the relationships that Part 2 detected, the Random Forest used its ensemble structure and feature-splitting mechanisms in order to learn the intricate relationships between alloy composition, thermal treatment, defect characteristics and lifespan.

3.3 Critical Review

A number of strengths were also indicated by the regression methodology, such as the application of a coherent preprocessing pipeline, relevant comparison of the baselines, and the systematic hyperparameter optimization of both ElasticNet and Random Forest. This guaranteed consistency in methodology across the models, as well as allowing significant interpretation of differences in performance. The presence of regularised and non-linear algorithms also gave equal consideration to simple and complex approaches.

There were certain limitations, however. Linear Regression and ElasticNet were not as objective in terms of offering opportunities to tune, and, therefore, limited the availability of performance improvement. The search space of ElasticNet can be considered relatively small and it might not have covered some configurations which would have allowed it to gain better regularisation control (Wang et al., 2025). In the case of the Random Forest, tuning did not consider other potentially significant parameters like max features, bootstrap, and tree-specific constraints, which may enhance the bias-variance tradeoff. In addition, no residual diagnostics or feature selection methodology was used that could have detected noise or redundant predictors.

4. Classification Implementation

4.1 Feature Crafting

In this case of the classification task, the goal is to see whether a manufactured metal part is at minimum safe lifespan as set by the company. The specification determines 1500 hours as the cut-off point beyond which a part is considered to be acceptable to be used operationally. On this need, one more binary target variable, safelabel1500, was developed whereby a value of 1 is assigned to all samples that have a lifetime of 1500 hours or more and the reverse is also true. This conversion generated 711 samples of the defective group and 289 samples of the safe group, which is a significant, though controllable, imbalance in the classes, as the number of values displays in the notebook.

Although the given threshold is a simple and understandable classification goal, it also brings about possible oversimplification. The spread of lifetime values in the dataset (359 to over 2000 hours) is apparent and some of the variables, such as Nickel%, Iron%, coolingRate, and categorical manufacturing characteristics, have complex and non-linear relationships with the lifespan (Dong et al., 2025). A breakdown into two binaries can then obscure subtle distinctions between divisions which are slightly above or below the 1500-hour point. Nonetheless, since the safety requirement of the company is practically relevant, and the specification permits binary classification, such a threshold is a defensible and operationally relevant classification.

4.2 Methodology

The classification methodology was developed to check whether a manufactured metal part is able to pass through the minimum safe lifespan threshold of the company. Based on the binary label that had been generated in the previous section 4.1, we chose two complementary families of models, namely, Logistic Regression and the Random Forest Classifier. Logistic Regression is an interpretable base that is able to represent a linear decision boundary. Owing to the fact that the Part 2 exploration showed a lack of linear correlations and evident interactions among features, it was complemented with equal weighting of classes in order to handle the moderate class imbalance. The Random Forest Classifier was, however, selected based on its capacity to represent non-linear associations, the variety of the types of features considered, and its resistance to noise - features that are certainly well-correlated with the complexity observed in the dataset.

There was an effective and uniform pre-processing pipeline. Lifespan feature was also eliminated to avoid trivial leakage, which directly gives the output label. All categorical variables that include parttype, microstructure, seedLocation and casttype were encoded by one-hot encoding, turning them into a format that is suitable for numeric and does not enforce any ordinal assumptions. Numerical variables were left in their original scale as tree-based models do not need standardisation, but the unified numeric-encoded dataset also served as an advantage of the Logistic Regression, as all the predictors were in comparable numeric values. An 80/20 train-test split was stratified to maintain the class distribution originally, in order to have a fair model generalisation.

The evaluation plan used threshold and probabilistic measures. The accuracy, precision, recall, and F1-score were calculated to give a balanced evaluation of the model behaviour among the imbalanced classes. Of particular interest were precision-recall trade-offs since it was more expensive to falsely declare defective parts as being safe. Also, ROC-AUC and PR-AUC measures were added to measure the ability to discriminate without the choice of a threshold.

4.3 Evaluation

The evaluation of the classification compared the Logistic Regression and the Random Forest Classifier with the binary label safelabel1500, the goal of which was to find out which model offers the most relevant distinction between safe and defective metal components. The assessments were all done on the held-out 20% test set to make sure that the evaluation was unbiased.

The Logistic Regression model gave a total accuracy of 0.65 with a distinct disparity in the predictive ability among the two classes. It was fairly good on defective parts (0.83) but had trouble recalling (0.63), thus giving it a low F1-score of 0.72. In the safe class, the recall was increased to 0.67, but the precision was reduced to 0.43, which shows a large number of false positives (Zhang et al., 2025). This inconsistency is indicative of the fact that the model only has a limited capability to represent the complex, non-linear relationships that are seen in the previous data exploration.

The Random Forest Classifier, in its turn though, performed much better in all measures. It obtained an accuracy of 0.89 and F1-score of 0.93 of the defective group. More importantly, it continued to show a balanced performance of the safe class with accuracy and recollection of 0.91 and 0.69 respectively. The results of these are indicative of the ability of

the model to capture interactions and feature hierarchies which are captured in a linear model. Stratification applied during the splitting process made sure that both classes were adequately represented during training, which also helped in creating good generalisation.

Random Forest outperformed probabilistic performance comparisons as well. There was a significant difference between the ROC curves: the AUC values of Logistic Regression were approximately 0.71 and the AUC of the Random Forest was about 0.97, which was almost near-perfect discrimination between classes.

```
from sklearn.metrics import roc_curve, auc

y_prob_lr = lr.predict_proba(X_test)[:,1]
y_prob_rf = rf.predict_proba(X_test)[:,1]

fpr_lr, tpr_lr, _ = roc_curve(y_test, y_prob_lr)
fpr_rf, tpr_rf, _ = roc_curve(y_test, y_prob_rf)

roc_auc_lr = auc(fpr_lr, tpr_lr)
roc_auc_rf = auc(fpr_rf, tpr_rf)

plt.figure(figsize=(8,6))
plt.plot(fpr_lr, tpr_lr, label=f'Logistic Regression (AUC={roc_auc_lr:.2f})')
plt.plot(fpr_rf, tpr_rf, label=f'Random Forest (AUC={roc_auc_rf:.2f})')
plt.plot([0,1],[0,1],'r--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve Comparison')
plt.legend()
plt.show()
```

Figure 5: ROC Curves for LR and RF (Colab Notebook File, p29)

In the same way, Precision-Recall curves demonstrated a significant advantage of the use of the Random Forest, where the average precision scores are about 0.93 as opposed to 0.46 with Logistic Regression. This is especially significant due to the skewed data, where PR curves can give a better informative insight into the positive-class performance (Hu et al., 2025).

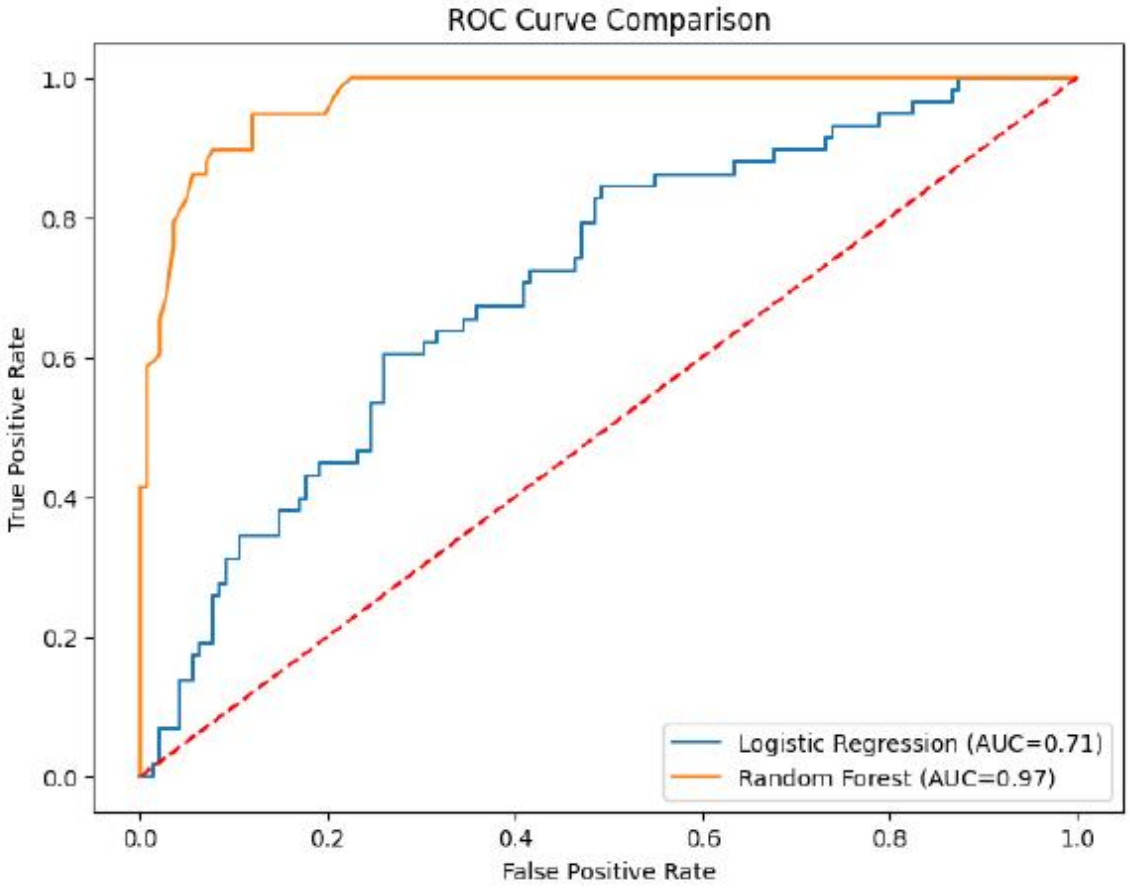


Figure 6: Precision-Recall curves for both models (Colab Notebook File, p30)

When the alloy composition variables (e.g., Nickel% and Iron %), number of defects, and processing parameters (e.g., coolingRate and forgeTime) were analyzed by the feature importance technique, they were all found to significantly contribute to the classification decision. These results are in line with what was observed in Part 2 and justify the non-linear nature of the dataset.

4.4 Critical Review

The classification approachology also displayed some strengths especially the right preprocessing, stratified splitting and balanced evaluation measures which extended beyond accuracy to incorporate ROC-AUC and Precision-Recall analysis. These options were crucial to the evaluation of model performance in the face of class imbalance and gave a clear picture of how the individual classifier dealt with bad and good parts. The good performance of the adopted method was also supported by the fact that the Random Forest made good use of features and has a strong predictive behaviour.

```
## Critical Review (Classification)
#Strengths: Proper label crafting, stratified split, balanced metrics, ROC/PR␣
 ↪curves.
#Weaknesses: Binary threshold may oversimplify, limited hyperparameter tuning␣
 ↪for LR.
# Future work: Try multi-class grouping, boosting methods, threshold␣
 ↪optimization based on business costs.
```

Figure 7: Critical Review (Colab Notebook File, p32)

Nonetheless, some restrictions were also apparent. The operationally justified binary threshold at 1500 hours simplifies the lifespan distribution and can superimpose valuable distinctions between performing parts in the moderate range. Hyperparameter tuning also had a limit especially in the case of Logistic Regression and random forest where wider grids or other optimisation techniques were possible to enhance the model performance.

5. Conclusions

The discussion showed that to predict the lifespan of metal parts, models should be used that are able to capture the complicated, non-linear association among thermal processing factors, alloy composition, and defect properties. The preliminary data analysis revealed that there were weak linear relationships and large dispersion in manufacturing conditions in the form of categories that may not be fully appropriate to basic linear techniques. This was actually proved by experiment: both Linear Regression and ElasticNet showed only minor improvements over the baseline. Conversely, Random Forest Regressor attained an R2 of 0.94, with much lower error rates, and have a high level of generalisation.

The Random Forest Classifier showed significantly better performance, having an ROC-AUC of approximately 0.97 and having balanced precision-recall behaviour, which is important to minimize the risk of false alarms raised by defective items.

Based on the general predictive power, the ability to interpret the importance of features and the ability to perform across tasks, the Random Forest model can be suggested as the only deployment solution. It offers stable lifespan prognostication and a stable pass/fail classification in helping make informed decisions in the manufacturing process.

6. References

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DECLARATION OF AI

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic - NO
- To create an outline of the topic - NO
- To explain concepts - YES
- To support my use of language - YES
- To summarise the following articles/resources - NO
- In other ways, as described below:
    1) Used AI to check and correct the formal academic style and tone of the report, ensuring consistent terminology and clear transitions between sections.